Anqi Tan

Ph.D. Student, Bioengineering · Seeking 2027 Internship in Diagnostics R&D

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Develops plasmon-enhanced lateral flow assays and the portable readers that quantify them, and validates both with human samples in lab and community settings. Available May–Aug 2027 (on-site or remote) and part-time remote during the school year.

EDUCATION

Ph.D. in Bioengineering, University of Illinois Urbana-Champaign Aug 2022 – Present
Advisor: Prof. Brian T. Cunningham · Nanosensors Group

B.S. in Mechanical Engineering (dual degree), UIUC & Zhejiang University Aug 2018 – Jun 2022

EXPERIENCE

Graduate Research Assistant, Nanosensors Group, UIUC Sep 2022 – Present

- Lead the Live Hormone project: developed a competitive plasmonic-fluor LFA for salivary cortisol (**LOD ≤0.12 ng/mL**, covering the physiological range) by optimizing antibody loading, blocking, and sample prep; collected saliva from **153 adults** in lab and community settings; **oral presentation, APHA 2026**.
- Designed and built a **~$100 pocket-size NIR fluorescence reader** (750 nm excitation, Raspberry Pi camera, **Android app in Kotlin**) that automatically quantifies test/control lines; imaging comparable to a benchtop ImageQuant scanner (first-author manuscript in preparation).
- Lead reader development for a **$4.87M DoD (PRMRP)** multi-site project with Stanford, WashU, and UCSF that builds a multiplexed IL-8/sTNFR-1 lateral flow test to classify sepsis/ARDS phenotypes at the bedside.
- Contributed to an ultrasensitive plasmon-enhanced LFA antigen test for *Neisseria gonorrhoeae* with WashU School of Medicine (*ACS Nano Medicine* 2026).

Research Assistant Intern, BrainCo (Zhejiang Qiangnao Technology), Hangzhou, China May 2022 – Aug 2022

- Supported two EEG studies at the company's brain-science institute: sleep assistance with acoustic and electrical stimulation, and fatigue monitoring with multimodal EEG.

Undergraduate Research Leader, ZJU-UIUC Institute 2019 – 2021

- Led a national-level undergraduate research project designing a re-localizable remote-center-of-motion mechanism for minimally invasive surgery (**first-author paper, IEEE ICARM 2021**).

SELECTED PROJECTS

CRISPR Mechanism Explainer (Ashby Prize Hackathon, NCSA) — 1st Place Oct 2025

- Two-person team; found the public benchmark was almost all "hit" labels, rebuilt a balanced validation set with experimental context, and fine-tuned an LLM (Qwen3) to classify CRISPR screen outcomes and explain the mechanism; shipped a Gradio chat demo.

PUBLICATIONS

First author: IEEE ICARM 2021 · pocket fluorescence reader manuscript (in preparation). **Co-author:** 4 journal papers (*Chemical Reviews*, *Lab on a Chip*, *ACS Nano Medicine*, *Biosensors*) and 1 Elsevier book chapter. **Talks:** APHA 2026 (oral), LOAC 2026.

HONORS & TEACHING

1st Place, Ashby Prize Computational Science Hackathon, NCSA (2025) · National Silver Medal, China "Internet+" Innovation & Entrepreneurship Competition (2021) · TA, BIOE/ECE 415 Biomedical Instrumentation Lab (2025) and ME 170 Computer-Aided Design (3 semesters)

SKILLS

Biosensing: lateral flow assays (competitive & sandwich), plasmonic-fluor labels, immunoassay optimization, ELISA, saliva/plasma sample prep, fluorescence imaging, human-sample studies

Engineering: SolidWorks, Creo, Fusion 360, 3D printing, laser cutting, optical/instrument design, COMSOL, FEA, design for assembly

Programming: Python, MATLAB, LabVIEW, Android (Kotlin), Raspberry Pi, Arduino, ESP32

Languages: English (fluent), Chinese (native)